Suggested problems: 2-4, 6-10

Problem 1: (Example 3.6.10) Suppose that a point X is chosen from the uniform distribution on $[0, 1]$, and after choosing a point $X = x$, a point Y is chosen from the uniform distribution on $[x, 1]$.

(a) Compute the marginal p.d.f. $f(y)$.

(b) Compute the conditional p.d.f. $g(x \mid y)$.

Problem 2: Suppose that either of two instruments might be chosen for making a certain measurement. Let Y be a random variable with $Y = 1$ or $Y = 2$, depending on whether the first or second instrument is chosen, with equal probability that either is chosen. If X is the value of the measurement, suppose that we have conditional p.d.f. functions

$$g(x \mid y = 1) = \begin{cases} 2x & : 0 < x < 1, \\ 0 & : \text{else}, \end{cases}$$

and

$$g(x \mid y = 2) = \begin{cases} 3x^2 & : 0 < x < 1, \\ 0 & : \text{else}. \end{cases}$$

(a) Compute the marginal p.d.f. $f(x)$.

(b) If the value of the measurement is $X = 1/4$, what is the probability that instrument 1 was used?